



FIRST PUBLIC RELEASE · BETA

What's in 2.5.0

One platform, **four tools**, one login.

CENTER FOR WATER RESILIENCY & RISK REDUCTION · FLORIDA ATLANTIC UNIVERSITY

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AT A GLANCE

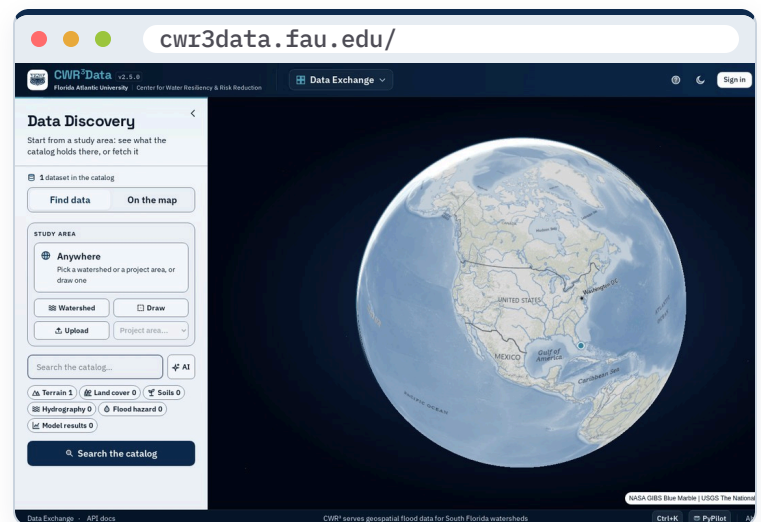


Fig. 1 — The home page: Data Discovery over a 3D globe, Florida pinned · cwr3data.fau.edu, v2.5.0

OPEN BETA · FREE TO TRY

Sign up with your **FAU email** at **cwr3data.fau.edu**.
Browsing the catalog is open to everyone.

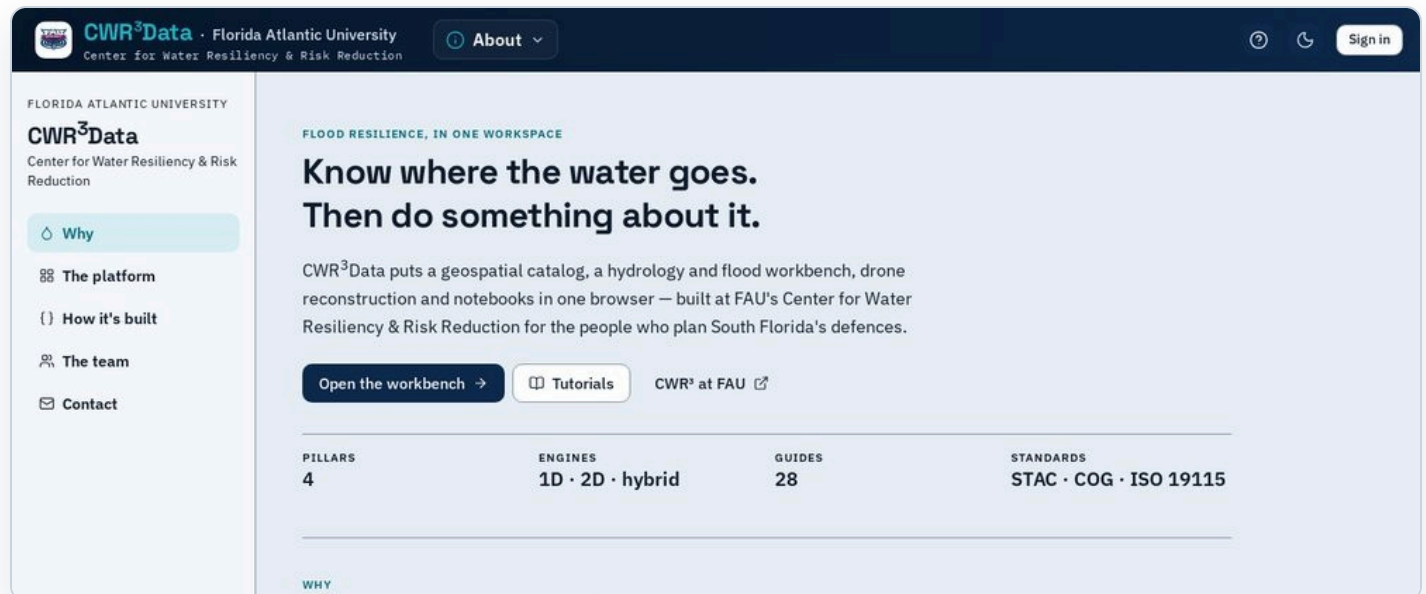
See it live at **GIS Mela** · Tue, Nov 17 · FAU Student Union

Flood data for South Florida — and the tools to use it

CWR³Data serves geospatial flood data for South Florida watersheds, and the tools to model, capture and share it.

What makes it worth an hour of your time is that the four tools are not four separate products. They share **one login, one project space, one catalog, and one API**. PyPilot, the built-in assistant, works inside each of them.

A delineated watershed can use rainfall from the catalog and terrain generated from a drone survey, while the results can be analyzed in a notebook with the project already mounted — without exporting a file or signing in again.



The home page: the pitch, the four pillars, and one button into the workbench · User Guide v3, p.27

4

tools in one workspace

1

login, project space,
catalog & API

28

click-by-click tutorials,
built in

0

lines of code required

You do not need to write code to use any of it. *If you want to, there is a notebook.*

What you get that a folder of desktop tools can't give you

Most flood work means four or five programs, a pile of downloads and a workstation to run them on. Here it is one browser tab, one project and one login.



Data and models in the same place

Search the catalog, preview a layer, and pull it straight into a model. Nothing is exported, re-projected and re-imported between programs.



Nothing to install; server-side GPU included

It runs in the browser. Two-dimensional rain-on-mesh simulations, photogrammetry workflows, and GeoAI jobs run on server-side GPU resources, not on your laptop.



Public data already wired in

NOAA Atlas 14 rainfall, USGS 3DEP LiDAR and stream gauges, NLCD land cover, NHDPlus flowlines, soils, and FEMA flood zones — fetched for your study area, not hunted down tile by tile.



Parameters estimated, not looked up

Curve number, time of concentration, and routing are derived for every subbasin. No gauge? LUNA, trained on about 2,100 gauged basins, can estimate them.



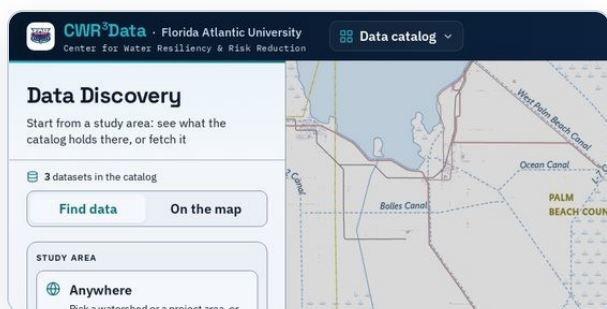
Drone surveys beside the hydrology

Photogrammetry, USGS LiDAR fusion, classification, and change detection live in the same workspace as the flood models — and export as ordinary COG and LAZ files.



An assistant that knows your project

PyPilot draws on the platform documentation and your project context, explains the result on your screen, and drafts notebook cells — and it acts only after you approve.



The platform menu at the top left of every page · User Guide v3, p.4

One menu, everywhere

The button at the top left names the tool you are in and opens onto the others, your projects, settings, the API reference and the tutorials. A tool your account cannot use yet is still listed, with the reason.

No large software installation or local data download is required — only a browser and an FAU email address.

The usual way, and the way it works here

STEP	TYPICAL TOOLCHAIN	CWR ³ DATA
Find the data	Hunt across agency portals; download zips by hand	Search one catalog over your study area and preview on the map
Get terrain	Download and mosaic DEMs or LiDAR tiles	3DEP elevation and LiDAR fetched for the area you drew
Set up a model	Desktop hydrology package; parameters selected or estimated manually	Delineate, then curve number (CN), time of concentration (Tc), and routing estimated per subbasin
Design storm	Look up Atlas 14 values and retype them	Atlas 14 depths pulled for your basin centroid
Run 2D flooding	A local solver and a workstation to feed it	Rain-on-mesh flooding simulated on a server-side GPU
Calibrate	Hundreds of model runs; ungauged basins set by judgment	In the example shown, LUNA reached a comparable optimum using about one-tenth as many model runs; it can also estimate without a gauge
Process a drone flight	A separate photogrammetry suite and license	AeroFuse in the same project, fused with public LiDAR
Analyze the output	Export files, open a local Python environment	Notebooks with your projects already mounted
Share it	Zip it and email it	Publish to the catalog with ISO metadata — open or by request

Open standards

STAC catalog, cloud-optimized GeoTIFF, LAZ point clouds, and ISO 19115 metadata — your results leave in formats any GIS reads.

Reproducible in the browser

The same project, data and parameters are there for anyone you share it with; no “works on my machine”.

Self-hostable

The platform can run on your own hardware if your data cannot leave it.

Find it, preview it, share it

Flood maps, elevation models, land cover, and hydrology results — searchable in plain English and previewed on the map before you download.



The catalog over Palm Beach and Broward, basemap menu open · User Guide v3, p.297

FILTER BY WHAT YOU NEED

Terrain

3DEP elevation and LiDAR

Land cover

NLCD classes for curve numbers

Soils

Hydrologic soil groups

Hydrography

NHDPlus flowlines and USGS gauges

Flood hazard

FEMA flood zones

Model results

Runs you and others publish

Keyword or plain language

Search by keyword, or switch on **AI** and describe the dataset you need — “flood zones near Fort Lauderdale”.

Preview first

Every match is drawn on the map; view any layer before you download it.

Four public basemaps

Topo, street, satellite, and imagery with labels — U.S. Government services, so no vendor tile limits.

Publish your own

Add datasets with full metadata — **open**, or **by request**.

From a study area to a download in three moves

1 Draw your study area

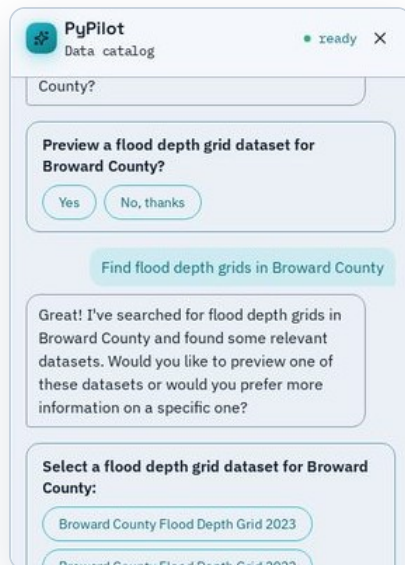
Draw a rectangle, select a watershed, or upload a boundary. The catalog filters to it.



Spatial extent set from a drawn rectangle over Palm Beach County · User Guide v3, p.294

2 Ask in plain English

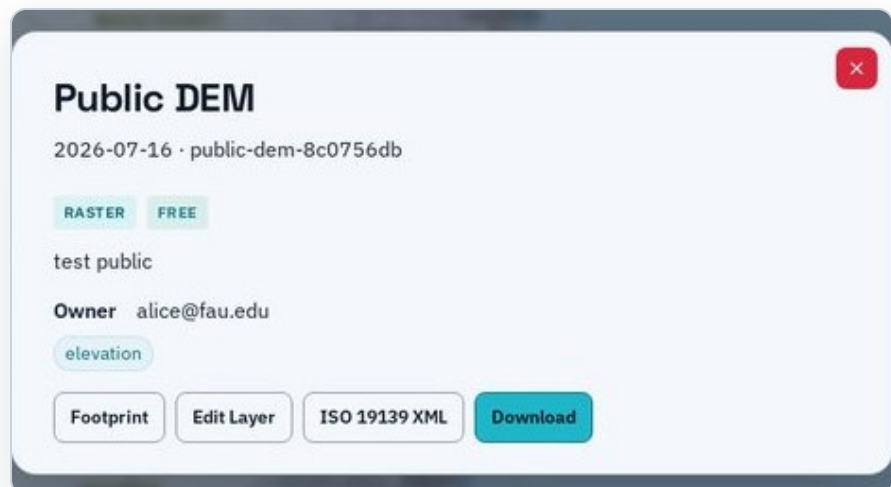
PyPilot searches the catalog for you and offers matching datasets to preview.



PyPilot finding flood-depth grids for Broward County · User Guide v2

3 Open a dataset

Metadata, badges, and actions in one card — footprint, ISO 19139 XML, download.



A dataset detail card · User Guide v3, p.295

TRY IT

Estimated time: 15 minutes

See what data exists for your area. 1 Draw a study area 2 Search

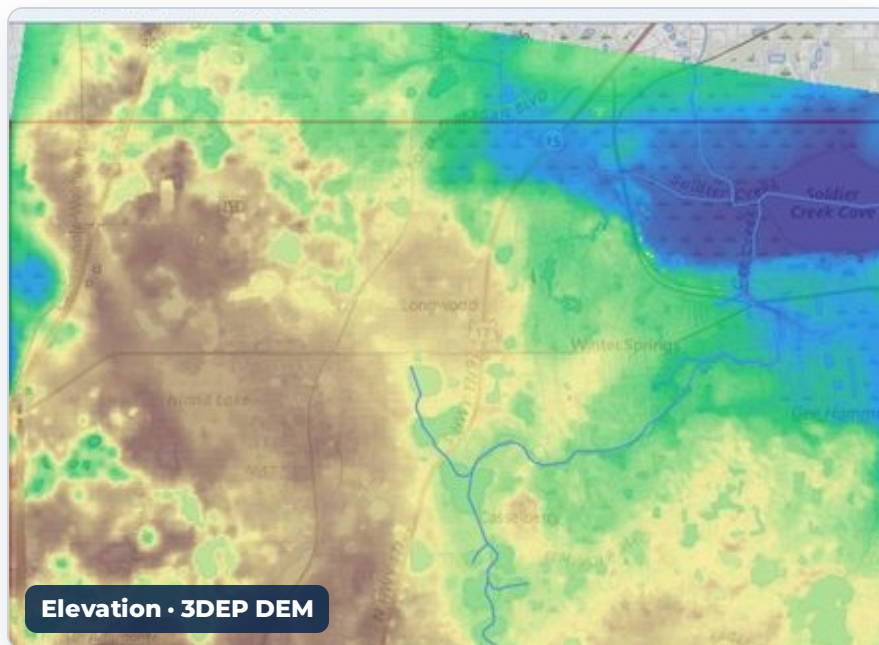
3 Preview a layer

The hydrology workbench

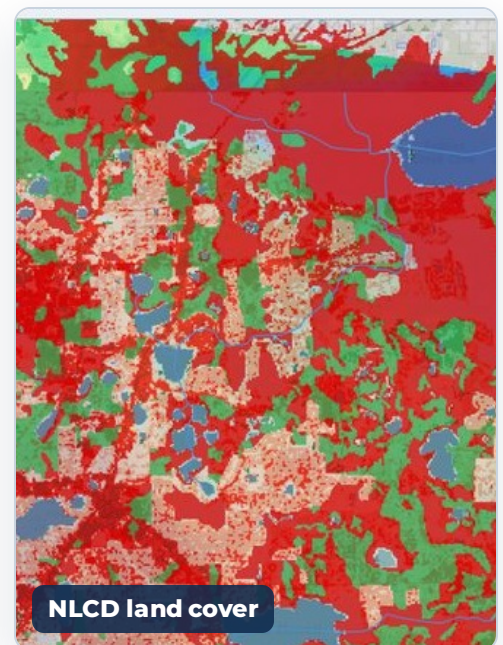
Delineate the watershed from elevation. The platform estimates curve number, time of concentration, and routing parameters, and pulls design rainfall from NOAA Atlas 14.

Identify Distance Area | Export Basemap

The map bar: identify, measure distance and area, export, switch the basemap · User Guide v3, p.235



3DEP terrain fetched for the study area · User Guide v3, p.41



NLCD land cover, the input to curve numbers · User Guide v3, p.76

Draw the area once. Terrain, land cover, soils, and NHDPlus flowlines are fetched for it, and every layer can be switched on, faded, or inspected from the map.

TRY IT

Estimated time: 1-2 hours

Run a flood model end to end. ① New project ② Draw an area ③ Delineate ④ Atlas 14 storm ⑤ Simulate

Rainfall Inputs
Gee Creek 10

DESIGN STORM

Duration
24 hours (1 day)

Return period
100-year

10.40" 24 hr this site
rainfall depth duration NOAA source

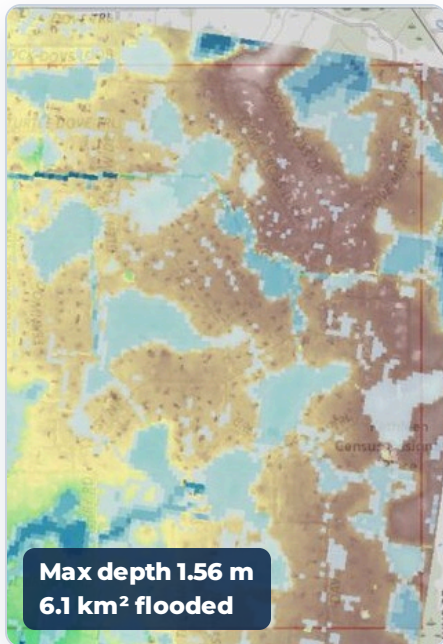
Depths come from NOAA Atlas 14 for your basin

Atlas 14 design storm: 24-hour, 100-year · User Guide v3, p.81

Three engines. Run it, check it, report it.

1D basin model

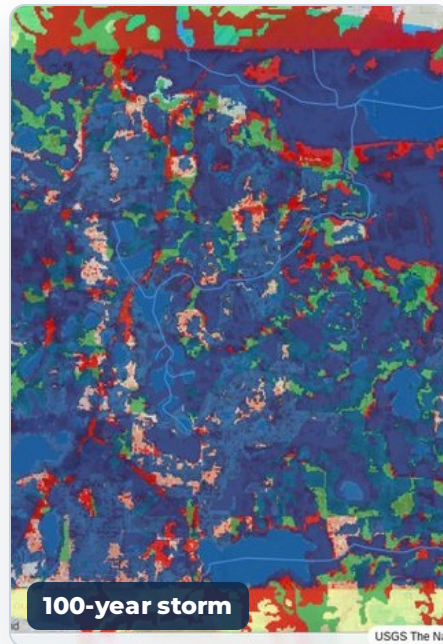
Fast runoff and routing through the basin.



Blackwater Creek 2D: flood depth after a GPU run · User Guide v3, p.138

2D rain-on-mesh

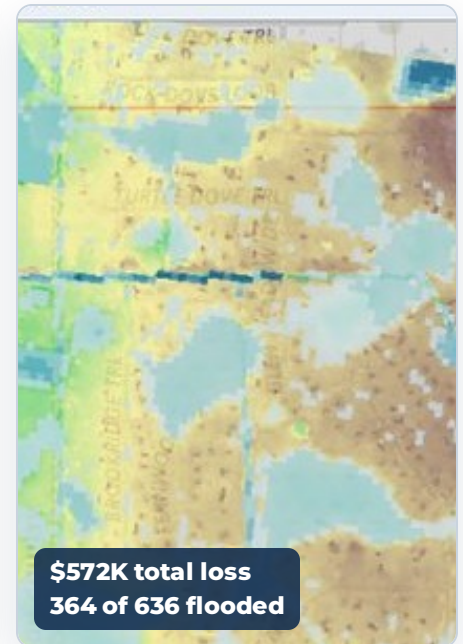
Rainfall is applied directly to the mesh; flooding is simulated on server-side GPUs.



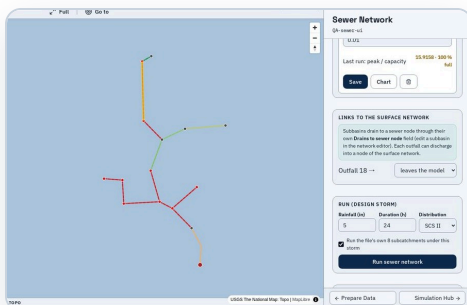
HAND inundation, 100-year storm, over land cover · User Guide v3, p.199

Coupled 1D + 2D

Channels and overland flow together.



Damage assessment on the same run · User Guide v3, p.235



A storm sewer after a run: pipes colored by fullness · User Guide v3, p.122

Storm sewers, and dual drainage

Import a SWMM **.inp** or draw the network; EPA SWMM 5.2 runs it under the design storm and colors each pipe by how full it ran. On a 2D mesh, manholes exchange flow between the sewer network and the surface domain. Generate flood maps, estimate monetary damages, and produce a report.

How damages are estimated. HAZUS Level-1 depth–damage curves (51 USACE occupancy classes) are applied to each structure in the USACE National Structure Inventory; losses are in U.S. dollars at [NSI valuation year]. The curves are national averages, so treat results as screening-level estimates, not appraisals · User Guide v3, pp. 230–238.

TRY IT

Estimated time: 1 hour

See the 2D flood mapping. ① 2D mesh project ② DEM ③
Configure the mesh ④ Run on the GPU

Calibrate with a gauge, or without one

LUNA (Learnable Unit Hydrograph Network Architecture) was trained on watershed attributes from about 2,100 gauged basins to estimate parameters for HEC-HMS, the Hydrologic Engineering Center's Hydrologic Modeling System.

Upstream catchments → Land cover • terrain • climate • soils → LUNA → HEC-HMS parameters

NO GAUGE Estimate the parameters

Most basins have no gauge, and calibration needs one. Click **Estimate Parameters**; on the tutorial project LUNA returned this table in about 30 seconds.

Curve number (CN) • initial abstraction ratio (Ia)	Soil Conservation Service (SCS) loss parameters
Clark Tc • Clark R (h)	time of concentration (Tc) and storage coefficient
Baseflow fraction • k	a linear-reservoir baseflow store
Suggested <code>cn_scale</code> • <code>lag_scale</code>	the two factors calibration adjusts; Apply writes them to the network
Distance from training basins	a percentile; at or above the 90th percentile the row turns amber — treat the values as a starting point

BENCHMARK • CASTRO VALLEY, JAN 1973

On HEC's Castro Valley validation example, the engine reached a Kling–Gupta efficiency (KGE) of **0.8120**, against **0.8107** for HEC's own calibration, a comparable result; both halves of a split-sample test rated satisfactory or better. Full record: PyWMP benchmark suite • User Guide v3, p. 216.

Conditions to state: HEC-HMS [version] • objective [] • [n] parameters and bounds • stopping criteria [] • hardware [] • seeds [].

HAVE A GAUGE Fit it by gradient

LUNA follows the gradient of the fit through a differentiable copy of your network; every trial is still scored by the real model. In the Gee Creek example, it reached a comparable objective value in 92 model evaluations, versus 931 with differential evolution:

Differential evolution	global search
	931
LUNA (gradient)	comparable objective value
92	

Results may vary by watershed, parameterization, and stopping criteria. Networks with Green-Ampt, Muskingum-Cunge, or a reservoir with outlet structures fall back to differential evolution, and the result says so.

No-gauge example. On Gee Creek, LUNA returned a curve number of 33 and flagged the basin at the 93rd percentile of distance from its training basins.

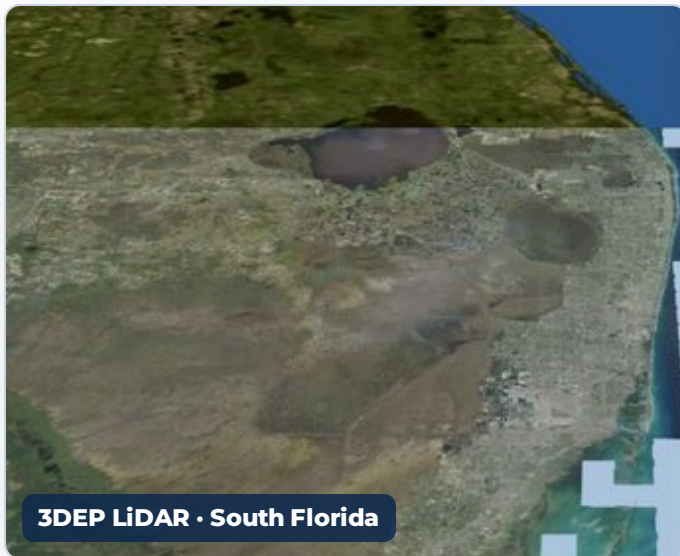
PERFORMANCE RATINGS

	NSE	KGE	PBIAS
Very good	> 0.80	> 0.85	< 5 %
Good	> 0.70	> 0.75	< 10 %
Satisfactory	> 0.50	> 0.50	< 15 %

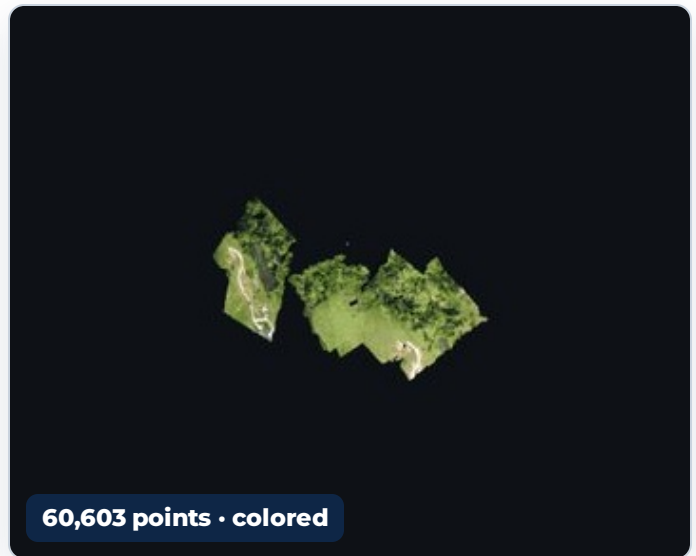
NSE: Nash–Sutcliffe efficiency • PBIAS: percent bias. Thresholds after Moriasi et al. (2007, 2015), as used in the platform.

Drone flights into usable terrain

Upload imagery; AeroFuse reconstructs georeferenced three-dimensional models, fuses public USGS LiDAR, and delivers surface models (DSM), bare-earth terrain models (DTM), and orthophotos.



Step 1: USGS 3DEP LiDAR fetched in the 3D scene · User Guide v3, p.312



Step 2: the fused drone model, Aukerman UAV · User Guide v3, p.316

One survey, four products

Every build delivers an **orthophoto**, a **surface model (DSM)**, a **bare-earth terrain model (DTM)** as cloud-optimized GeoTIFFs, and the **point cloud** as LAZ — georeferenced and ready for any GIS. Publish them to the catalog, or feed the DTM straight into a PyWMP-WEB model in the same project.

Clean in the browser

Tidy the point cloud without leaving the page.

Label a few, find the rest

Label a small set of example features, and machine learning identifies similar features.

Fly twice, measure the change

AeroFuse identifies and quantifies changes between two surveys.

How the DTM is made. With **Classify ground points** on, ground points are classified automatically to build a first bare-earth model. Review and correct the classes by hand in the 3D editor; the learned classifier improves from your corrections · User Guide v3, pp. 309, 316.

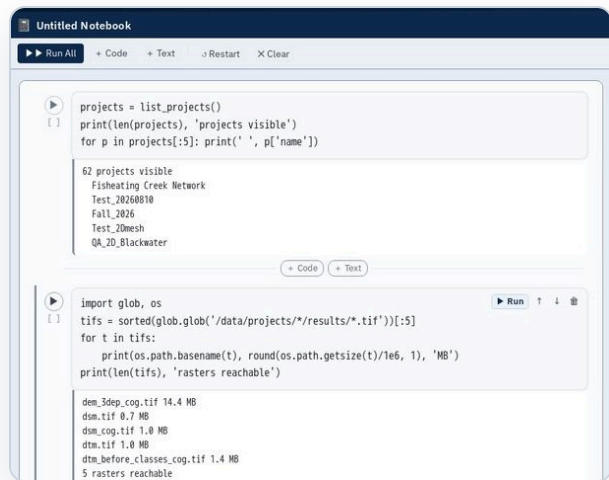
TRY IT

Estimated time: 1-2 hours

Turn drone imagery into terrain. ① New project ② Upload images or pull 3DEP LiDAR ③ Build

Use code, or work through the assistant

Server-side Python with your projects already mounted — and PyPilot, an assistant that knows where you are and what you are looking at.



```

projects = list_projects()
print(len(projects), 'projects visible')
for p in projects[:5]: print(' ', p['name'])

62 projects visible
Fisheating Creek Network
Test_20260810
Fall_2026
Test_20mesh
QA_2D_Blackwater

import glob, os
tifs = sorted(glob.glob('/data/projects/*/results/*.tif'))[:5]
for t in tifs:
    print(os.path.basename(t), round(os.path.getsize(t)/1e6, 1), 'MB')
print(len(tifs), 'rasters reachable')

dem_3dep_cog.tif 14.4 MB
dsm.tif 8.7 MB
dsm_cog.tif 1.8 MB
dtm.tif 1.0 MB
dtm_before_classes_cog.tif 1.4 MB
5 rasters reachable
  
```

A notebook listing projects and reading their rasters · User Guide v2

Notebooks

No downloading and re-uploading: your projects are mounted. Launch GeoAI jobs as server GPU tasks (they queue when the GPU is busy) —

Segmentation

Building detection

Tree detection

Change detection

— with the supplied pre-trained detectors or models you train yourself. Supported imagery resolution: [confirm]. Or describe what you want and the assistant drafts runnable cells.

TRY IT

Estimated time: 30 minutes

Work in code. ① Notebooks ② New notebook from a template

PyPilot, across all four tools

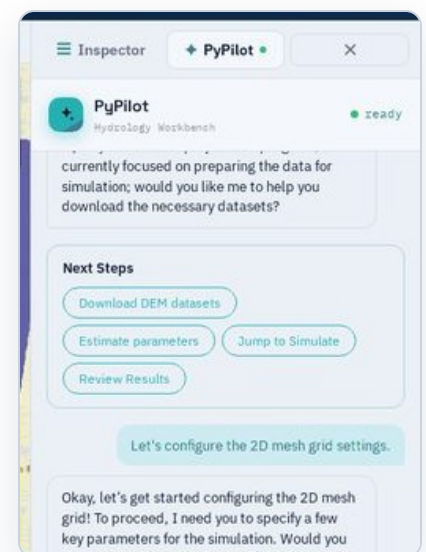
PyPilot uses the platform documentation and your current project context to generate responses, explains the results in front of you, drafts reports — and **carries out actions only after you approve them.**

Review AI output. Verify technical recommendations, AI-generated code, and interpretations before you rely on them.

TRY IT

Estimated time: 10 minutes

Test the assistant. ① Press **Ctrl+K** anywhere ② Ask about your project



PyPilot proposing next steps · User Guide v2

Find your way, then pick one

Take the tour first — it takes about a minute. Then pick the row that matches your work; each workflow can be completed as a focused introductory exercise.

Ctrl+K from anywhere

Jump to any tool, page, or project — or type a question; answers draw on the platform documentation.

A one-minute tour

It walks the menu on your first sign-in. Skip the tour or replay it from the Help (?) button.

28 tutorials, built in

Click-by-click guides with screenshots — each gives the manual path first, then the phrasing to ask the assistant instead.

IF YOU WANT TO...	TRY THIS	EST. TIME
See what data exists for your area	Data catalog: ① Draw a study area ② Search ③ Preview a layer	15 minutes
Run a flood model end to end	PyWMP-WEB: ① New project ② Delineate ③ Atlas 14 storm ④ Simulate	1–2 hours
See the 2D flood mapping	PyWMP-WEB: ① 2D mesh project ② DEM ③ Configure the mesh ④ Run on the GPU	1 hour
Turn drone imagery into terrain	AeroFuse: ① New project ② Upload images or pull 3DEP LiDAR ③ Build	1–2 hours
Work in code	Notebooks: ① New notebook from a template	30 minutes
Test the assistant	Any page: ① Press Ctrl+K ② Ask about your project in plain language	10 minutes

Test what matters to your expertise

If you are a hydrologist, critically evaluate the parameter estimates. If you fly drones, bring a flight that gave other software trouble. We would rather hear that a number looks wrong than that the buttons work.

Built on public data

- **NOAA Atlas 14** design rainfall, pulled straight into models
- **USGS stream gauges** to calibrate against real flows
- **USGS 3DEP LiDAR** fused with your drone flights
- Flood maps, elevation, land cover and hydrology for **South Florida watersheds**

Three steps to your account

1

Sign up

Use your **FAU email** (@fau.edu or a subdomain).

2

Tell us

An administrator approves each account; automated notifications are not yet enabled. Email **[support email]** after signing up.

3

Take the tour

About one minute; then open the tool that matches your work.

Browsing the catalog is open to everyone. You only need an account to download data, run models and keep projects.

Tell us what you find

Use **Report an issue** in the bar at the bottom of any page or in the user menu.

What a report records. Your description, the page it was sent from, your account, and up to six attachments (screenshots, PDF, or DOCX, 15 MB each). Reports go to the CWR³ team and are visible to platform administrators · User Guide v3, pp. 28, 354.

The beta platform incorporates research software that is undergoing continued testing and refinement. Reports of unexpected behavior are especially valuable during beta testing.

About the Images

Screenshots are reproduced from the *CWR³Data User Guide* (v3, September 2026; v2 where marked); the corresponding page is provided in each caption.

SEE IT LIVE • WORLD GIS DAY 2026

Try it with us at GIS Mela

Tuesday, November 17 · 9 AM – 3 PM · FAU Student Union, Boca Raton

Bring a study area, a drone flight or a question — the CWR³Data table will run it with you.



CWR³Data cwr3data.fau.edu

SCAN TO RSVP



FREE • ALL MAJORS